

Problem 10 – Smart ATM Banking System

Difficulty: ★★☆☆ Beginner+

Overview

Build a Streamlit ATM Banking Application that simulates the basic features of an ATM. The application should allow customers to securely log in, check their balance, deposit and withdraw money, change their PIN, and view their account information through a clean graphical interface.

Project Requirements

1. Create the Bank Account

Store the customer's information in a dictionary.

Information	Value
Account Number	1001
Customer Name	Ahmed Ali
PIN	1234
Balance	5000 EGP

2. Customer Login

Before accessing the ATM, the customer must enter:

- Account Number
- PIN

Validate both values before allowing access to the system.

If either value is incorrect, display an error message.

3. ATM Operations

After a successful login, allow the customer to perform the following operations:

- Check Balance
 - Deposit Money
 - Withdraw Money
 - Change PIN
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4. Check Balance

Display:

- **Customer Name**
 - **Account Number**
 - **Current Balance**
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5. Deposit Money

Ask the customer to enter a deposit amount.

Validation:

- **Amount must be greater than 0.**

If valid:

- **Update the balance.**
 - **Display a success message.**
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6. Withdraw Money

Ask the customer to enter a withdrawal amount.

Validation:

- **Amount must be greater than 0.**
- **Amount cannot exceed the available balance.**

If the withdrawal is successful:

- **Update the balance.**
- **Display a confirmation message.**

Otherwise, display an appropriate error message.

7. Change PIN

Ask the customer to enter:

- Current PIN
- New PIN
- Confirm New PIN

Validation:

- Current PIN must be correct.
- New PIN must contain exactly 4 digits.
- Confirmation must match the new PIN.

If all validations pass:

- Update the PIN.
- Display a success message.

8. Logout

Provide a button that logs the customer out and returns to the login screen.

Bonus ★

Create a Transaction History.

Store every transaction in a list of dictionaries.

Each transaction should contain:

- Transaction Type
- Amount
- Date
- Balance After Transaction

Example:

Type	Amount	Date	Balance
Deposit	500 EGP	06/07/2026	5500 EGP
Withdrawal	300 EGP	06/07/2026	5200 EGP

Streamlit Requirements

Required Text Functions

Use all of the following:

- `st.title()`
 - `st.header()`
 - `st.subheader()`
 - `st.write()`
 - `st.code()`
 - `st.divider()`
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Required Input Widgets

Use:

- `st.text_input()` → Account Number
 - `st.text_input(type="password")` → PIN
 - `st.selectbox()` → ATM Operation
 - `st.number_input()` → Transaction Amount
 - `st.text_input()` → New PIN
 - `st.button()` → Login
 - `st.button()` → Perform Operation
 - `st.button()` → Logout
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Required Status Elements

Use at least five of the following:

- `st.info()`
- `st.success()`
- `st.warning()`
- `st.error()`
- `st.spinner()`

Suggested uses:

- ATM instructions
 - Successful transactions
 - Large withdrawal warning (more than 3000 EGP)
 - Incorrect login or insufficient balance
 - Transaction processing animation
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Layout Requirements

Sidebar

Display:

- ATM Services
 - Current Date
 - Banking Tips
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Tabs

Create four tabs.



Tab 1 – Login

Allow the customer to log in.

Display:

- **Account Number**
- **PIN**
- **Login Button**

If the login is successful, allow access to the remaining tabs.

Tab 2 – ATM Operations

This tab is used to perform banking operations.

Use two columns.

Left Column

Display:

- Operation Selector
- Amount Input (when required)
- New PIN Inputs (for Change PIN)
- Perform Operation Button

Right Column

Display:

- Current Balance
 - Last Transaction
 - Transaction Status
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Tab 3 – Account Information

Display the customer's account details.

Show:

- Customer Name
 - Account Number
 - Current Balance
 - Total Deposits (*Bonus*)
 - Total Withdrawals (*Bonus*)
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Tab 4 – Transaction History (Bonus)

Display all recorded transactions.

For each transaction show:

- Transaction Type
- Amount
- Date
- Balance After Transaction

Display the newest transaction first.

Expected Interface

 Smart ATM Banking System

Sidebar

ATM Services

Current Date

Banking Tips

Tabs

[ Login] [ ATM Operations] [ Account Information] [ Transaction History]

Account Number

[1001]

PIN

[****]

[Login]

Operation

[Deposit ▼]

Amount

[500]

[Perform Operation]

Current Balance

5500 EGP

Last Transaction

Deposit: 500 EGP

 Transaction Completed Successfully!

Challenge ★★ ★

Support multiple bank accounts.

Store at least 5 customer accounts in a list of dictionaries.

After logging in:

- **Each customer should only access their own account.**
- **All operations should update only that customer's data.**

At the end of the day, display:

- **Total Deposits**
 - **Total Withdrawals**
 - **Total Transactions**
 - **Customer with the Highest Balance**
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Suggested Functions

Organize your program into reusable functions.

Examples:

- **login()**
 - **check_balance()**
 - **deposit_money()**
 - **withdraw_money()**
 - **change_pin()**
 - **record_transaction()**
 - **display_history()**
 - **logout()**
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Python Topics Covered

- **Variables**
 - **Strings**
 - **Integers**
 - **Floats**
 - **Lists**
 - **Dictionaries**
 - **Arithmetic Operators**
 - **Comparison Operators**
 - **Assignment Operators**
 - **Membership Operators**
 - **if, elif, else**
 - **Functions**
 - **datetime.today()**
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Streamlit Topics Covered

Text

- **st.title()**
- **st.header()**
- **st.subheader()**
- **st.write()**
- **st.code()**
- **st.divider()**

Input Widgets

- **st.text_input()**
- **st.number_input()**
- **st.selectbox()**
- **st.button()**

Status Elements

- `st.info()`
- `st.success()`
- `st.warning()`
- `st.error()`
- `st.spinner()`

Layout

- `st.sidebar`
- `st.tabs()`
- `st.columns()`
- `st.session_state`

Learning Objectives

After completing this project, you should be able to:

- Store customer data using dictionaries.
- Build a secure login system.
- Validate user input before processing transactions.
- Update account balances safely.
- Record transaction history using a list of dictionaries.
- Work with dates using the datetime module.
- Manage application state with `st.session_state`.
- Build a clean, multi-tab Streamlit application using sidebars, columns, and status messages.